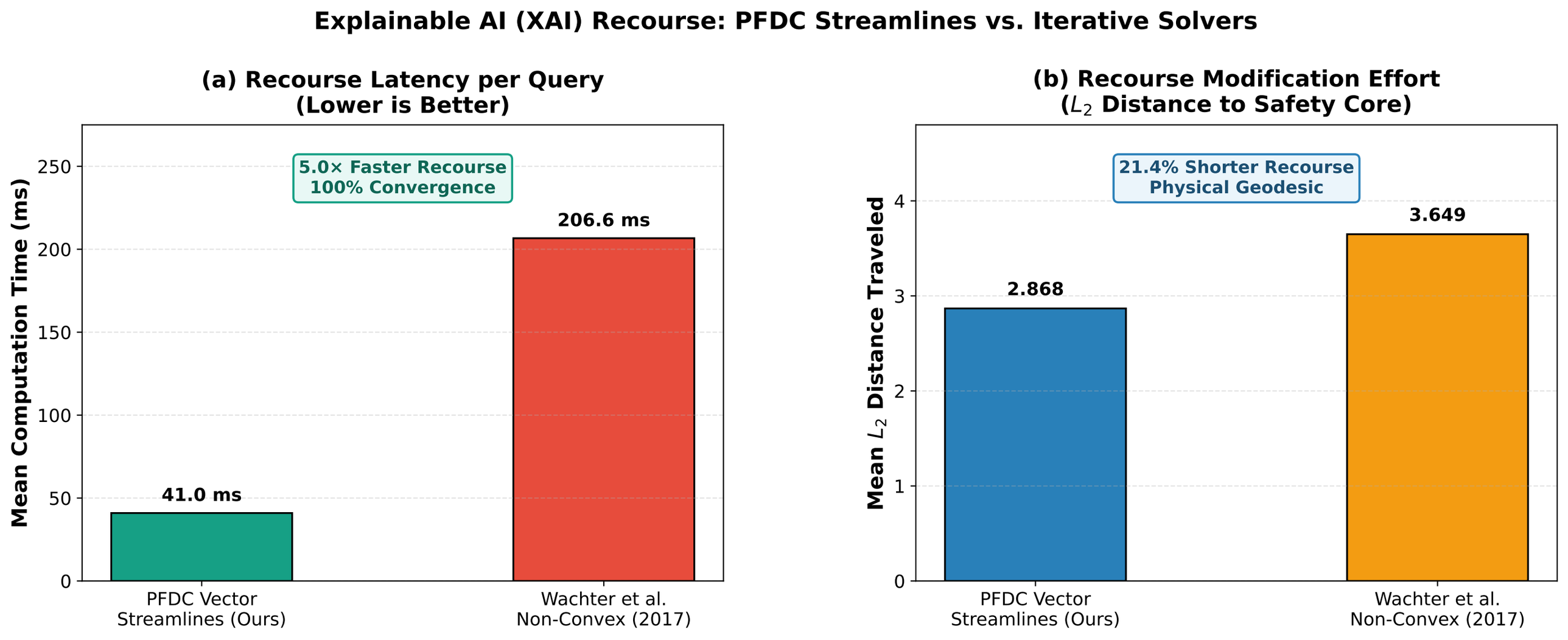
Explainable AI (XAI) Recourse: PFDC Streamlines vs. Iterative Solvers
(a) Recourse Latency per Query
(Lower is Better)
Mean Computation Time (ms)
0
50
100
150
200
250
5.0× Faster Recourse
100% Convergence
41.0 ms
206.6 ms
PFDC Vector
Streamlines (Ours)
Wachter et al.
Non-Convex (2017)
(b) Recourse Modification Effort
(L_2 Distance to Safety Core)
Mean L_2 Distance Traveled
0
1
2
3
4
21.4% Shorter Recourse
Physical Geodesic
2.868
3.649
PFDC Vector
Streamlines (Ours)
Wachter et al.
Non-Convex (2017)